

Ring Theory

Ring Theory Notes

We will discuss primality in the Gaussian integers. Recall that the Gaussian integers are given by

$$\begin{aligned}\mathbb{Z}[i] &= \{a + bi \mid a, b \in \mathbb{Z}\} \\ &\cong \frac{\mathbb{Z}[x]}{(x^2 + 1)}.\end{aligned}$$

The Gaussian integers are equipped with a norm $N: \mathbb{Z}[i] \rightarrow \mathbb{Z}_{\geq 0}$ given by $N(a + bi) = (a + bi)(a - bi) = a^2 + b^2$, which is a multiplicative norm satisfying $N(zw) = N(z)N(w)$. Furthermore, with this norm, the Gaussian integers form a Euclidean domain. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Lemma: Let $q \in \mathbb{Z}[i]$. Then, there is a prime integer $p \in \mathbb{Z}$ such that $N(q) = p$ or $N(q) = p^2$.

Proof. Since q is not a unit, $N(q) \neq 1$. Therefore, $N(q)$ is a nontrivial product of prime integers, and since q is prime in $\mathbb{Z}[i]$, q must divide one of the prime integer factors of $N(q)$. Let p be this prime integer.

Then, $q|p$ in $\mathbb{Z}[i]$, so $N(q)|N(p) = p^2$, so $N(q) = p$ or $N(q) = p^2$. $\square$

Example: We observe that 3 is a prime element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is a UFD, so irreducibles are prime, meaning we only need to show that 3 is irreducible in $\mathbb{Z}[i]$.

Since $N(3) = 9$, it follows that the norm of any factor of 3 must be a divisor of 9. Therefore, it follows that the norm of any divisor of 3 is equal to 1, 3, or 9. If the norm is 9, then the factor is an associate of 3, while if the norm is 1, then the factor is a unit, so it follows that any nontrivial factor of 3 would have norm equal to 3. Yet, there are no such elements.

Meanwhile, 5 is not a prime element of $\mathbb{Z}[i]$, since we may write $5 = (2 + i)(2 - i)$.

We will now focus on proving one of the most celebrated theorems in number theory: Fermat's Theorem on sums of squares. We will say a prime integer $p \in \mathbb{Z}$ *splits* in $\mathbb{Z}[i]$ if it is not a prime element of $\mathbb{Z}[i]$. For example, 5 splits in $\mathbb{Z}[i]$ while 3 does not.

Lemma: A positive prime integer $p \in \mathbb{Z}$ splits in $\mathbb{Z}[i]$ if and only if it is the sum of two squares in $\mathbb{Z}$.

Proof. Suppose $p = a^2 + b^2$ with $a, b \in \mathbb{Z}$. Then,

$$p = (a + bi)(a - bi)$$

in $\mathbb{Z}[i]$, so since $N(a \pm bi) = a^2 + b^2 = p \neq 1$, it follows that neither of the factors is a unit in $\mathbb{Z}[i]$, so that p is not irreducible, hence not prime.

Conversely, suppose p is not irreducible in $\mathbb{Z}[i]$. Then, p has an irreducible factor $q \in \mathbb{Z}[i]$ that is not an associate of p . Since $q|p$, multiplicativity of the norm gives that $N(q)|N(p) = p^2$, hence $N(q) = p$ as q and p are not associates and q is not a unit. If we have $q = a + bi$, then

$$\begin{aligned}p &= N(q) \\ &= a^2 + b^2.\end{aligned}$$

$\square$

Next, we discuss a characterization of splitting via a single congruence.

Lemma: A positive odd prime integer p splits in $\mathbb{Z}[i]$ if and only if it is congruent to 1 modulo 4.

Proof. Our goal is to understand when $\mathbb{Z}[i]/(p)$ is an integral domain. Yet, we have the series of isomorphisms

$$\begin{aligned}\frac{\mathbb{Z}[i]}{(p)} &\cong \frac{\mathbb{Z}[x]/(x^2 + 1)}{(p)} \\ &\cong \frac{\mathbb{Z}[x]}{(p, x^2 + 1)}\end{aligned}$$

$$\begin{aligned} &\cong \frac{\mathbb{Z}[x]/(p)}{(x^2 + 1)} \\ &\cong \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2 + 1)}. \end{aligned}$$

In particular, this means that p splits in $\mathbb{Z}[i]$ if and only if $x^2 + 1$ has a root in $\mathbb{Z}/p\mathbb{Z}$, or that there is some n such that $n^2 \equiv -1$ modulo p . We will show that this condition is equivalent to p being congruent to 1 modulo 4.

Let $G = (\mathbb{Z}/p\mathbb{Z})^\times$, and fix a generator g . Observe that p is odd, so $p - 1 = 2\ell$ for some integer ℓ . Let $\bar{n}$ denote the class of an integer $n \bmod (p)$. For any $n \notin (p)$, it follows that there is some m such that $\bar{n} = g^m$.

Observe that the class $\bar{-1}$ generates the unique subgroup of order 2 of G , and since g^ℓ does the same, it follows that $g^\ell = -1$. In particular, we have that $n^2 = -1$ modulo p if and only if $g^{2m} = g^\ell$, or that $2m \equiv \ell \bmod 2\ell$, or that $(p - 1) = 2\ell$ is a multiple of 4, or $p \equiv 1 \bmod 4$. $\square$

This gives the following.

Theorem (Fermat's Theorem on Sums of Squares): A positive odd prime $p \in \mathbb{Z}$ is a sum of two squares if and only if $p \equiv 1 \bmod 4$.

Field/Galois Theory

Field Theory Notes

Field Theory Worked Solutions

Problem (September '84, Problem 7): Factor $x^8 - 1$ into irreducibles in $\mathbb{Z}/p\mathbb{Z}[x]$ for all primes p .

Solution: We start by understanding the case where $p = 2$. Then, we observe that $x^8 - 1 = x^{2^3} - 1$, so that $x^8 - 1 = (x - 1)^{2^3}$.

We start by observing that over $\mathbb{Z}$, $x^8 - 1$ factors into irreducibles given by

$$x^8 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)\Phi_8(x),$$

where $\Phi_n(x)$ denotes the n th cyclotomic polynomial. In particular, we observe that these cyclotomic polynomials are given by

$$\begin{aligned} \Phi_1(x) &= x - 1 \\ \Phi_2(x) &= x + 1 \\ \Phi_4(x) &= x^2 + 1 \\ \Phi_8(x) &= x^4 + 1. \end{aligned}$$

Therefore, we must understand the factorization into irreducibles over $\mathbb{Z}/p\mathbb{Z}$ of $x^2 + 1$ and $x^4 + 1$. When p is congruent to 1 modulo 4, then -1 is a square modulo p , so we may factor $x^2 + 1 = (x - a)(x + a)$. When p is not congruent to 1 modulo 4, then -1 is not a square modulo p , meaning that $x^2 + 1$ is irreducible.

Now, we observe that the square of every odd number is congruent to 1 modulo 8, so that $x^8 - 1$ divides $x^{p^2-1} - 1$, so we have that $x^4 + 1$ factors completely over $\mathbb{F}_{p^2}$. In particular, we may factor $x^4 + 1$ into the product of the minimal polynomials of the roots of $x^4 + 1$ in $\mathbb{F}_{p^2}$ over $\mathbb{F}_p$.

Problem (August '99, Problem 9): If p is a prime number congruent to 1 mod 8, show that 2 is a quadratic residue mod p ; i.e., there is an integer a such that $a^2 \equiv 2$ modulo p .

Solution: First, we show that there exists some α such that $\alpha^4 = -1$. For this, it suffices to show that the polynomial $x^4 + 1$ is not irreducible over $\mathbb{F}_p$. Since $\mathbb{F}_p$ is equal to the roots of the polynomial $p(x) = x^p - x$,

we observe then that since $8|p-1$, it follows that x^8-1 divides $x^p-x = x(x^{p-1}-1)$. Therefore, we have $x^4+1|p(x)$. Since $p(x)$ splits over $\mathbb{F}_p$, it follows that x^4+1 splits over $\mathbb{F}_p$, so that x^4+1 has a root in $\mathbb{F}_p$.

Consider now the element $a = \alpha + \alpha^{-1}$. Then, we have

$$\begin{aligned} a^2 &= (\alpha + \alpha^{-1})^2 \\ &= \alpha^2 + \alpha^{-2} + 2 \\ &= 2 \end{aligned}$$

mod p , since $\alpha^{-2} = -\alpha^2$.

Problem (August '21, Problem 7):

- (a) Let $n \in \mathbb{N}$, and let F be an algebraically closed field of characteristic 0 or of characteristic $p > 0$ where $p \nmid n$. Prove that F contains a primitive n th root of unity.
- (b) According to (a), $\overline{\mathbb{F}_3}$, the algebraic closure of the field of three elements contains a 13th root of unity. Call it ω . Compute $[\mathbb{F}_3(\omega) : \mathbb{F}_3]$.

Solution:

- (a) If the characteristic of F equals zero, the smallest subfield of F containing 1 is isomorphic to $\mathbb{Q}$, so we will assume without loss of generality that $\mathbb{Q} \subseteq F$. The n th cyclotomic polynomial, $\Phi_n(x)$, then has $\text{spl}_{\mathbb{Q}}(\Phi_n(x)) \subseteq F$ a separable extension that contains a primitive n th root of unity. Therefore, F has a primitive n th root of unity.

If the characteristic of F equals p , then we observe that the polynomial $p(x) = x^n - 1$ over $\mathbb{F}_p$ has $p'(x) = nx^{n-1} \neq 0$, whence $p(x)$ splits fully into linear factors in the algebraic closure F as $p(x)$ is then separable. There is then some extension $\mathbb{F}_{p^d}$ of $\mathbb{F}_p$ containing all the roots of $p(x)$, and in particular, we have $n|d-1$. Since any element of $(\mathbb{F}_{p^d})^\times$ is a root of unity, it follows that there is some primitive $(d-1)$ th root of unity contained in $\mathbb{F}_{p^d}$, whence there is also a primitive n th root of unity.

- (b) If ω is a primitive 13th root of unity, then ω is a root of the polynomial $p(x) = x^{13} - 1$. We observe that over $\mathbb{Z}$, hence over $\mathbb{F}_3$, we have that $p(x)|x^{26} - 1$, whence we have $p(x)$ divides $q(x) = x^{27} - x$. In particular, this means that $p(x)$ splits completely in $\mathbb{F}_{p^3} \subseteq F$ and does not split over $q_2(x) = x^9 - x$, so that $[\mathbb{F}_3(\omega) : \mathbb{F}_3] = 3$.

Problem (August '99, Problem 8): If F is finite, show that every element $\alpha \in F$ is the sum $\alpha = \beta_1^2 + \beta_2^2$.

Solution: Since F is finite, we have that $F = \mathbb{F}_{p^n}$ for some prime p and some $n \geq 1$.

If $p = 2$, then the Frobenius map is given by squaring, and since F is finite, the map is surjective, so for any $\alpha \in \mathbb{F}_{2^n}$, there is some $\beta \in \mathbb{F}_{2^n}$ such that $\alpha = \beta^2$.

If $p \neq 2$, we consider the sets

$$\begin{aligned} A &= \{\beta^2 \mid \beta \in F\} \\ B &= \{\alpha - \zeta \mid \zeta \in A\}. \end{aligned}$$

Observe that $(\mathbb{F}_{p^n})^\times$ is cyclic, so $|A| = \frac{p^n+1}{2}$, and similarly, we have $|B| = \frac{p^n+1}{2}$. Yet, this means that $A \cap B \neq \emptyset$, so that there are some $\beta_1^2, \beta_2^2 \in A$ such that $\alpha - \beta_1^2 = \beta_2^2$, whence $\alpha = \beta_1^2 + \beta_2^2$.

Problem (August '08, Problem 3): Prove or disprove the following statements about irreducibility.

- (a) If $\text{Gal}(E/\mathbb{Q}) = S_n$ for E the splitting field over $\mathbb{Q}$ of a polynomial $f(x) \in \mathbb{Q}[x]$ of degree n , then $f(x)$ must be irreducible.
- (b) If α is algebraic over $\mathbb{Q}$, then its minimal polynomial $\mu_{\alpha, \mathbb{Q}}(x)$ must be irreducible.

(c) If $A \in \text{Mat}_n(\mathbb{R})$, then its minimal polynomial $\mu_A(x)$ must be irreducible.

Solution:

(a) To start, we have that $\text{Gal}(E/\mathbb{Q})$ acts transitively on the roots of $f(x)$; since there are at most n roots for $f(x)$ in its splitting field, it follows that $\text{Gal}(E/\mathbb{Q}) \hookrightarrow S_m$, where m is the number of distinct roots of $f(x)$ in E . Yet, since $\text{Gal}(E/\mathbb{Q}) = S_n$, it follows that $f(x)$ has n distinct roots in E , whence it is separable.

Since $f(x)$ is a separable polynomial whose Galois group is S_n , its Galois group is a transitive subgroup of S_n , whence $f(x)$ is irreducible.

Problem (Algebra II HW 11, Problem 6): Let p be a prime, $K = \mathbb{F}_p(t)$, the field of rational functions in one variable over $\mathbb{F}_p$. Let $\sigma: K \rightarrow K$ be the unique automorphism of K such that $\sigma(t) = t + 1$, and $G = \langle \sigma \rangle$. Observe that G is cyclic of order p . Let $F = K^G$. Find an explicit element s such that $F = \mathbb{F}_p(s)$, and prove your answer.

Solution: Consider the product given by

$$s = \prod_{i=0}^{p-1} (t + i),$$

so that

$$\begin{aligned} \sigma(s) &= \prod_{i=0}^{p-1} (\sigma(t) + i) \\ &= \prod_{i=0}^{p-1} (t + i + 1) \\ &= s \end{aligned}$$

since σ has order p . Therefore, $s \in K^G$, so that $\mathbb{F}_p(s) \subseteq K^G$.

Consider now the tower of extensions given by $\mathbb{F}_p(s) \subseteq K^G \subseteq K$. We may write

$$K = (\mathbb{F}_p(s))(t),$$

so we may compute

$$\begin{aligned} [K : \mathbb{F}_p(s)] &= [\mathbb{F}_p(s)(t) : \mathbb{F}_p(s)] \\ &\leq p, \end{aligned}$$

where the bound appears from the fact that t is a root of the polynomial

$$\prod_{i=0}^{p-1} (x + i) - s \in \mathbb{F}_p(s)[x].$$

Meanwhile, by Artin's Lemma, we know that K/K^G is Galois with Galois group G , whence $[K : K^G] = |G| = p$. Hence, we have $[K^G : \mathbb{F}_p(s)] = 1$, so that $\mathbb{F}_p(s) = K^G$.

Problem (Algebra II HW 11, Problem 1): Let p be a prime, n a positive integer, and let $\Phi_n(x) = x^{p^n} - x$ be an element of $\mathbb{F}_p[x]$. Prove that Φ_n is equal to the product of all the monic irreducible polynomials in $\mathbb{F}_p[x]$ whose degrees divide n , with each polynomial appearing with multiplicity one.

Solution: First, we observe that if $\overline{\mathbb{F}_p}$ is an algebraic closure, then the splitting field of $\Phi_n(x)$ is given by $\mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$. We may write

$$\Phi_n(x) = \prod_{\alpha \in \mathbb{F}_{p^n}} (x - \alpha)$$

split in $\overline{\mathbb{F}_p}$. Then, any irreducible factor f_i of Φ_n will define a subfield of the splitting field of $\Phi_n(x)$, which is given by $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ for some n . We have that $m|n$, so that $\deg(f_i)|n$ and that

$$f_i(x) = \prod_{\alpha \in \mathbb{F}_{p^m}} (x - \alpha)$$

fully splits in $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$. In particular, this means that $f_i(x)$ has multiplicity one in the product decomposition of $\Phi_n(x)$.

Similarly, if $f_m(x)$ is a monic irreducible polynomial in $\mathbb{F}_p[x]$ with degree $m|n$, then $f_m(x)$ defines a field extension $\mathbb{F}_p \subseteq F \subseteq \mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$ with $[F : \mathbb{F}_p] = m$, whence $F = \mathbb{F}_{p^m}$. Since $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$, it follows that every root of f_m in $\overline{\mathbb{F}_p}$ is a root in $\Phi_n(x)$, whence $f_m(x)|\Phi_n(x)$, and $f_m(x)$ has multiplicity one by the earlier discussion. Therefore,

$$\Phi_n(x) = \prod_{m|n} f_m(x),$$

where $f_m(x)$ is a monic irreducible factor of degree m for $\Phi_n(x)$ and each f_m appears with multiplicity one.

Problem (January '20, Problem 8): Let K/F be a field extension. Suppose that $K = F(a, b)$ for some $a, b \in K$ such that $a^2 \in F$ and $b^2 \in F$.

- (a) Prove that $[K : F] \leq 4$.
- (b) Give an example (with full proof) where $[K : F] = 4$.
- (c) Assume that $\text{char}(F) \neq 2$. Prove that the extension K/F is Galois.
- (d) Now, assume that F is finite. Prove that $[K : F] \leq 2$.

Solution:

- (a) We have

$$[F(a, b) : F] = [F(a, b) : F(b)][F(b) : F].$$

Observe that $\mu_{a, F(b)} | \mu_{a, F}$, with $\mu_{a, F}$ and $\mu_{b, F}$ having degree 2. Therefore, we have

$$\begin{aligned} [F(a, b) : F] &= [F(a, b) : F(b)][F(b) : F] \\ &\leq 4. \end{aligned}$$

- (b) We consider the case of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We start by noting that 2 and 3 are coprime. Observe that if $a + b\sqrt{2} = \sqrt{3}$ with $a, b \in \mathbb{Q}$, then we get

$$3 = a^2 + 2b^2 + 2ab\sqrt{2},$$

giving that $\sqrt{2}$ is rational, which is a contradiction. Therefore, we have that $\sqrt{3}$ is not contained in $\mathbb{Q}(\sqrt{2})$, whence the minimal polynomial is still degree 2. Therefore, we get

$$\begin{aligned} [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] &= [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] \\ &= 4. \end{aligned}$$

- (c) We observe that a and b are roots of the separable polynomials $x^2 - a^2$ and $x^2 - b^2$ respectively. We have that $F(a, b) = F(a)F(b)$ can be expressed as a compositum, and that $F(a)/F$ and $F(b)/F$ are separable extensions of degree 2, hence they are Galois extensions. Since the compositum of two Galois extensions is Galois, we get that K/F is Galois.
- (d) If F is finite, then $F = \mathbb{F}_q$ for some $q = p^m$. Since $F(a)/F$ and $F(b)/F$ are field extensions of degree at most 2, the result follows if either $F(a) = F$ or $F(b) = F$. Else, if both $F(a)/F$ and $F(b)/F$ are extensions of degree 2, then by uniqueness of finite fields, they are both equal to the field $\mathbb{F}_{q^2}$, whence $[F(a, b) : F] \leq 2$.

Galois Theory Notes

General Galois Theory Facts

Theorem: If $f(x) \in K[x]$ is a separable polynomial of degree n , then f is irreducible if and only if the Galois group $\text{Gal}(\text{spl}_K(f(x))/K)$ is a transitive subgroup of S_n .

Proof. If $f(x)$ is irreducible, then we know that the Galois group acts transitively on the roots.

Meanwhile, if $f(x)$ is reducible, then $f(x)$ is a product of distinct irreducibles since it is separable. Let r_i and r_j be roots of different irreducible factors of $f(x)$. Then, for each $\sigma \in \text{Gal}(\text{spl}_K(f(x))/K)$, we have $\sigma(r_i)$ has the same minimal polynomial over K as r_i , so we cannot have $\sigma(r_i) = r_j$, whence the Galois group is not transitive. $\square$

Galois Groups of Polynomials

Definition: Let $x_1, \dots, x_n$ be indeterminates. The *elementary symmetric functions* $s_1, \dots, s_n$ are defined by

$$\begin{aligned} s_1 &= x_1 + x_2 + \dots + x_n \\ s_2 &= \prod_{i < j} x_i x_j \\ &\vdots \\ s_n &= x_1 x_2 \dots x_n. \end{aligned}$$

The *general polynomial* of degree n is the polynomial

$$g(x) = (x - x_1)(x - x_2) \dots (x - x_n)$$

with roots $x_1, \dots, x_n$.

Observe that the general polynomial can be written as

$$(x - x_1)(x - x_2) \dots (x - x_n) = x^n - s_1 x^{n-1} + \dots + (-1)^n s_n.$$

This means that if F is some field, then the extension $F(x_1, \dots, x_n)$ is a Galois extension of the field $F(s_1, \dots, s_n)$ since it is the splitting field of the general polynomial of degree n . Furthermore, $F(s_1, \dots, s_n)$ is contained in the fixed field of the action of S_n on $F(x_1, \dots, x_n)$, where $\sigma \cdot x_i = x_{\sigma(i)}$ for any $\sigma \in S_n$. Since the fixed field of S_n has index $n!$ in $F(x_1, \dots, x_n)$ by the fundamental theorem of Galois theory, it follows that we have

$$[F(x_1, \dots, x_n) : F(s_1, \dots, s_n)] \leq n!,$$

whence $F(s_1, \dots, s_n)$ is the full fixed field of $F(x_1, \dots, x_n)$.

Theorem: The general polynomial

$$x^n - s_1 x^{n-1} + s_2 x^{n-2} + \dots + (-1)^n s_n$$

over the field $F(s_1, \dots, s_n)$ is separable with Galois group S_n .

Now, we will discuss the discriminant, which is an essential tool in understanding the Galois groups of polynomials, especially those of degrees two, three, and four.

Definition: The *discriminant* of $x_1, \dots, x_n$ is given by

$$\Delta = \prod_{i < j} (x_i - x_j)^2.$$

The discriminant of a polynomial is the discriminant of the roots of the polynomial.

We start by observing that the discriminant is a symmetric function in $x_1, \dots, x_n$, so it is an element of the field $K = F(s_1, \dots, s_n)$.

Recalling the definition of the alternating group A_n , we have that $\sigma \in S_n$ is in A_n if and only if σ fixes

$$\sqrt{\Delta} = \prod_{i < j} (x_i - x_j)$$

in $\mathbb{Z}[x_1, \dots, x_n]$. Notice then that if F has characteristic not equal to 2, then $\sqrt{\Delta}$ generates the fixed field of A_n , which is a quadratic extension of K .

If $f(x)$ is a monic degree n polynomial with roots $\alpha_1, \dots, \alpha_n$, then the discriminant is given by

$$\Delta = \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

Observe that $\Delta \in F$ since Δ is fixed by the Galois group. Furthermore, $\Delta = 0$ if and only if $f(x)$ is separable, so if F is perfect (i.e., characteristic zero or finite), then $f(x)$ is reducible if and only if $\Delta = 0$ since every irreducible polynomial over a perfect field is separable. Additionally, observe that $\sqrt{\Delta} \in F$ if and only if the Galois group of $f(x)$ is contained in A_n .

Now, we start our investigation with the polynomials of degree 2. The discriminant of a polynomial $x^2 + ax + b$ is given by $\Delta = (\alpha - \beta)^2$, which can be written as

$$\begin{aligned} \Delta &= s_1^2 - 4s_2 \\ &= a^2 - 4b. \end{aligned}$$

The polynomial is separable if and only if $a^2 - 4b \neq 0$, and the Galois group is a subgroup of $S_2 = \mathbb{Z}/2\mathbb{Z}$. This means that the Galois group is trivial if and only if $a^2 - 4b$ is a rational square.

Next, we consider the cubic polynomials. If we have

$$f(x) = x^3 + ax^2 + bx + c,$$

which upon the substitution $x = y - a/3$, yields a reduced cubic

$$g(y) = y^3 + py + q,$$

where

$$\begin{aligned} p &= \frac{1}{3}(3b - a^2) \\ q &= \frac{1}{27}(2a^3 - 9ab + 27c). \end{aligned}$$

Since the roots of the polynomials f and g differ by the constant $a/3$, and the formula for the discriminant involves the difference of these roots, it follows that the polynomials have the same discriminant.

Letting α, β, γ be the roots of the polynomial $g(y)$, then we have

$$\begin{aligned} g'(y) &= (y - \alpha)(y - \beta) + (y - \alpha)(y - \gamma) + (y - \beta)(y - \gamma) \\ g'(\alpha) &= (\alpha - \beta)(\alpha - \gamma) \end{aligned}$$

$$\begin{aligned} g'(\beta) &= (\beta - \alpha)(\beta - \gamma) \\ g'(\gamma) &= (\gamma - \alpha)(\gamma - \beta). \end{aligned}$$

Therefore, we have

$$\begin{aligned} \Delta &= ((\alpha - \beta)(\alpha - \gamma)(\beta - \gamma))^2 \\ &= -g'(\alpha)g'(\beta)g'(\gamma). \end{aligned}$$

Since $g'(y) = 3y^2 + p$, it follows that

$$\begin{aligned} -\Delta &= (3\alpha^2 + p)(3\beta^2 + p)(3\gamma^2 + p) \\ &= 27\alpha^2\beta^2\gamma^2 + 9p(\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2) + 3p^2(\alpha^2 + \beta^2 + \gamma^2) + p^3. \end{aligned}$$

These expressions are elementary symmetric functions of the roots, where for g , $s_1 = 0$, $s_2 = p$, and $s_3 = -q$, whence

$$\Delta = -4p^3 - 27q^2.$$

Therefore, expressing Δ in terms of a, b, c , we have

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc.$$

This allows us to compute the Galois group as follows.

- (i) If $f(x)$ is reducible, then either $f(x)$ splits into three linear factors or into a linear factor and an irreducible quadratic. In the first case, the Galois group is trivial and in the second case, the Galois group is of order 2.
- (ii) If $f(x)$ is irreducible, then a root of $f(x)$ generates a degree 3 extension over F . Therefore, the degree of the splitting field over F is divisible by 3.

Since the Galois group is a subgroup of S_3 , it follows that there are only two options, either A_3 or S_3 . In particular, the Galois group is A_3 if and only if the discriminant Δ is a square.

If Δ is a square, then the splitting field of the irreducible cubic $f(x)$ is obtained by adjoining any root of $f(x)$ to F . If Δ is not the square of an element of F , then the splitting field of $f(x)$ is of degree 6 over F , given by $F(\theta, \sqrt{\Delta})$ where θ is some root of $f(x)$. This extension is Galois over F generated by the automorphisms σ and τ , where σ takes θ to any other root of $f(x)$ fixing $\sqrt{\Delta}$ and τ takes $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ and fixing θ .

Finally, we discuss the Galois group of a quartic polynomial

$$f(x) = x^4 + ax^3 + bx^2 + cx + d.$$

Then, by substituting with $x = y - a/4$, this yields the quartic

$$g(y) = y^4 + py^2 + qy + r$$

with

$$\begin{aligned} p &= \frac{1}{8}(-3a^2 + 8b) \\ q &= \frac{1}{8}(a^3 - 4ab - 8c) \\ r &= \frac{1}{256}(-3a^4 + 16a^2b - 64ac + 256d). \end{aligned}$$

Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of $g(y)$ and G the Galois group for the splitting field of $g(y)$.

If $g(y)$ is reducible, then the following cases hold.

- (i) If $g(y)$ splits into a linear and a cubic, then G is the Galois group of the cubic.
- (ii) If $g(y)$ splits into two irreducible quadratics, then the splitting field is $F(\sqrt{\Delta_1}, \sqrt{\Delta_2})$, where Δ_1 and Δ_2 are the discriminants of the quadratics.
 If Δ_1 and Δ_2 do not differ by a square, then G is isomorphic to a Klein 4-subgroup of S_4 .
 If Δ_1 is a square multiplied by Δ_2 , then $F(\sqrt{\Delta_1}, \sqrt{\Delta_2}) = F(\sqrt{\Delta_1})$ is a quadratic extension with G isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

Now, we have the case of $g(y)$ irreducible. Then, the Galois group is transitive on its roots. There are five transitive subgroups (up to conjugation) of S_4 , given by copies of S_4, A_4, D_4, V, C , where V denotes the Klein 4-group and C denotes the copy of $\mathbb{Z}/4\mathbb{Z}$.

We consider the elements

$$\begin{aligned}\theta_1 &= (\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4) \\ \theta_2 &= (\alpha_1 + \alpha_3)(\alpha_2 + \alpha_4) \\ \theta_3 &= (\alpha_1 + \alpha_4)(\alpha_2 + \alpha_3)\end{aligned}$$

inside the splitting field for $g(y)$. Notice that the elements are permuted among themselves by the permutations of S_4 . The stabilizer of θ_1 is the dihedral group D_4 , while the stabilizer of all three of θ_1, θ_2 , and θ_3 is the copy of V .

The elementary symmetric functions in the θ_i are fixed by S_4 , so they are in F . These elementary symmetric functions evaluated on $\theta_1, \dots, \theta_3$ are given by $2p, p^2 - 4r$, and $-q^2$. This gives that θ_1, θ_2 , and θ_3 are the roots of

$$h(x) = x^3 - 2px^2 + (p^2 - 4r)x + q^2.$$

This is known as the *resolvent cubic* for the quartic $g(y)$. It can be shown (see DF p.614) that the discriminant Δ of the resolvent cubic is equal to the resolvent of $g(y)$. In particular, this gives

$$\Delta = 16p^4r - 4p^3q^2 - 128p^2r^2 + 144pq^2r - 27q^4 + 256r^3.$$

Remark: In the special case where $g(y) = y^4 + py^2 + r$ (which emerges during analysis of quartics whose roots are nested square roots), we have

$$\Delta = 16p^4r - 128p^2r^2 + 256r^3.$$

In particular, since the splitting field for the resolvent cubic is a subfield of the splitting field of the original quartic, the Galois group for $h(x)$ is a quotient of G , so knowing the action of the Galois group of $h(x)$ on the roots of $h(x)$ gives information about the Galois group of $g(y)$.

- (i) If the resolvent cubic is irreducible, and Δ is not a square, then G is not contained in A_4 with the Galois group of the resolvent cubic equal to S_3 . In particular, this means that the degree of the splitting field for $g(y)$ is divisible by 6, so that $G = S_4$.
- (ii) If the resolvent cubic is irreducible, and Δ is a square, then G is a subgroup of A_4 and 3 divides the order of G since the Galois group of the resolvent cubic is A_3 , so that $G = A_4$.
- (iii) If the resolvent cubic is reducible, and $h(x)$ splits completely, then every element of G fixes the elements θ_1, θ_2 , and θ_3 , so $G = V$.
- (iv) If the resolvent cubic splits into a linear factor and a quadratic, then G stabilizes θ_1 but not θ_2 or θ_3 , so $G \subseteq D_4$ and $G \not\subseteq V$, so $G = D_4$ or $G = C$.

Now, notice that $F(\sqrt{\Delta})$ is the fixed field of the elements of $G \cap A_4$. We have $D_4 \cap A_4 = V$ and $C \cap A_4 = \{e, (1, 3)(2, 4)\}$, the latter of which is not transitive. Therefore, we have the case $D_4 \cap A_4 = V$ if and only if $g(y)$ is irreducible over $F(\sqrt{\Delta})$.

Galois Theory Worked Solutions

Problem (August '89, Problem 8): Let $\beta = \sqrt{2 + \sqrt{2}}$. Show $\mathbb{Q}(\beta)/\mathbb{Q}$ is Galois with G cyclic; set up the Galois correspondence.

Solution: Working over the algebraic numbers, we observe that $\sqrt{2 + \sqrt{2}}$ is a root of the irreducible polynomial

$$f(x) = x^4 - 4x^2 + 2.$$

The discriminant is given by

$$\begin{aligned}\Delta &= 16(-4)^4(2) - 128(-4)^2(2)^2 + 256(2)^3 \\ &= 2^{11},\end{aligned}$$

so that $\mathbb{Q}(\sqrt{\Delta}) = \mathbb{Q}(\sqrt{2})$. Then, over $\mathbb{Q}(\sqrt{2})$, we have

$$f(x) = ((x^2 - 2) - \sqrt{2})((x^2 - 2) + \sqrt{2}),$$

which means that $f(x)$ is not irreducible over $\mathbb{Q}(\sqrt{2})$. In particular, this means that the Galois group is equal to $\mathbb{Z}/4\mathbb{Z}$, and in particular, since the degree of $\mathbb{Q}(\beta)$ over $\mathbb{Q}$ is 4, it follows that the extension is indeed Galois.

Problem (January '17, Problem 1): Consider the polynomial $f(x) = x^4 - 2x^2 - 6$. Prove this polynomial is irreducible. Describe the splitting field of this polynomial over $\mathbb{Q}$, and the Galois group of this splitting field.

Solution: We observe that this polynomial is contained in $\mathbb{Z}[x]$, so viewing $f(x)$ as an element of $\mathbb{Z}[x]$, we have that $f(x)$ is monic and $6 \notin (2^2)$, where $(2) \subseteq \mathbb{Z}$ is the ideal generated by the prime 2. By Eisenstein's criterion, it follows that $f(x)$ is irreducible over $\mathbb{Z}$, and by Gauss's Lemma, $f(x)$ is irreducible over $\mathbb{Q}$. Let K denote the splitting field of $f(x)$ over $\mathbb{Q}$.

Considering the set of roots

$$W = \left\{ \sqrt{1 + \sqrt{7}}, -\sqrt{1 + \sqrt{7}}, \sqrt{1 - \sqrt{7}}, -\sqrt{1 - \sqrt{7}} \right\},$$

we observe that if we let

$$L = \mathbb{Q}(\sqrt{1 + \sqrt{7}}),$$

then $f(x)$ is reducible over L , but has only two roots in L , so that $[K : L] = 2$. Since $[L : \mathbb{Q}] = 4$, it follows that $[K : \mathbb{Q}] = 8$.

Since $f(x)$ is irreducible, it follows that the Galois group of $f(x)$ acts transitively on the roots of f , and has order 8, so that $G \cong D_8$.

Problem (August '02, Problem 9): Construct a Galois extension of $\mathbb{Q}$ of degree 3.

Solution: Consider ζ_7 , a primitive 7th root of unity over $\mathbb{Q}$. Then, we have that $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ is a Galois extension, since $\mathbb{Q}(\zeta_7)$ is the splitting field of the separable polynomial $x^7 - 1$, and has $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$. A Galois extension of degree 3 corresponds to a subgroup of index 3 in $(\mathbb{Z}/7\mathbb{Z})^\times$, which is the unique subgroup of order 2 in $(\mathbb{Z}/7\mathbb{Z})^\times$. If we let $\langle \sigma \rangle \subseteq \text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ be this subgroup, then σ is given by an element of order 2 in $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$; since complex conjugation is a $\mathbb{Q}$ -automorphism of $\mathbb{Q}(\zeta_7)$, it follows that σ is complex conjugation.

Observe now that

$$\sum_{i=0}^1 \sigma^i(\zeta_7) = \zeta_7 + \zeta_7^{-1}$$

is invariant under σ , so that $\mathbb{Q}(\zeta_7 + \zeta_7^{-1}) \subseteq \mathbb{Q}(\zeta_7)^\sigma$. Furthermore, letting $\alpha = \zeta_7 + \zeta_7^{-1}$, we have

$$\zeta_7^2 - \alpha\zeta_7 + 1 = 0,$$

whence $\mu_{\zeta_7, \mathbb{Q}(\alpha)}(x)$ has degree at most 2.

Therefore, we observe that $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ is an extension with $[\mathbb{Q}(\zeta_7) : \mathbb{Q}(\zeta_7 + \zeta_7^{-1})] = 2$, whence $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ is the desired extension of degree 3 over $\mathbb{Q}$.

Problem (August '08, Problem 8):

- (a) Find a splitting field E of the polynomial $x^4 + 3x^3 + 4x^2 + 3x + 3$ over $F = \mathbb{Q}$, and find its degree $[E : F]$.
- (b) Find the Galois group $\text{Gal}(E/F)$ of the extension field.
- (c) Diagram the lattice of subgroups of the Galois group and the corresponding lattice of intermediate subfields of E/F .

Solution:

- (a) Taking the factorization over $\mathbb{Q}$, we have that the polynomial factorizes as $(x^2 + 1)(x^2 + 3x + 3)$. Over $\mathbb{C}$, we may factorize this as

$$p(x) = (x - i)(x + i)\left(x - \left(-\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)\right)\left(x - \left(-\frac{3}{2} - i\frac{\sqrt{3}}{2}\right)\right).$$

Adjoining the roots to $\mathbb{Q}$ gives that $E \subseteq \mathbb{Q}(i, \sqrt{3}) \subseteq E$. Therefore, $E = \mathbb{Q}(i, \sqrt{3})$. We have that

$$\begin{aligned} [E : \mathbb{Q}] &= [\mathbb{Q}(i, \sqrt{3}) : \mathbb{Q}(i)][\mathbb{Q}(i) : \mathbb{Q}] \\ &= 4, \end{aligned}$$

where we use the fact that $\mu_{\sqrt{3}, \mathbb{Q}(i)} = x^2 - 3$ is irreducible over $\mathbb{Q}[i]$ as 3 is not a square mod 4.

- (b) We claim that the maps

$$\begin{aligned} \sigma &= \begin{cases} i \mapsto -i \\ \sqrt{3} \mapsto \sqrt{3} \end{cases} \\ \tau &= \begin{cases} i \mapsto i \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases} \end{aligned}$$

are automorphisms of $\mathbb{Q}(i, \sqrt{3})/\mathbb{Q}$.

Problem (August '20, Problem 8): Let K be the splitting field of the polynomial $f(x) = (x^3 - 11)(x^2 + x - 1)$ over $\mathbb{Q}$.

- (a) Find (with proof) the degree of the extension $K/\mathbb{Q}$.
- (b) Determine the isomorphism class of the Galois group $\text{Gal}(K/\mathbb{Q})$, and prove your answer.

Solution:

- (a) Considering the embedding of $\mathbb{Q}[x] \subseteq \mathbb{C}[x]$, we observe that the roots of $f(x)$ are given by

$$A = \left\{ \sqrt[3]{11}, \omega \sqrt[3]{11}, \omega^2 \sqrt[3]{11}, \frac{-1 + \sqrt{5}}{2}, \frac{-1 - \sqrt{5}}{2} \right\},$$

where ω is a primitive cube root of unity. In particular, this means that the splitting field K of $f(x)$ is contained in $F = \mathbb{Q}(\omega, \sqrt[3]{11}, \sqrt{5})$. Meanwhile, we have $F \subseteq K$ since we may find $\omega = \frac{\omega \sqrt[3]{11}}{\sqrt[3]{11}}$ and $\sqrt{5} = 1 + 2\frac{-1 + \sqrt{5}}{2}$, so that $K = \mathbb{Q}(\omega, \sqrt[3]{11}, \sqrt{5})$.

To compute the degree, we observe that the minimal polynomials are given by

$$\begin{aligned} \mu_{\sqrt[3]{11}, \mathbb{Q}}(x) &= x^3 - 11 \\ \mu_{\omega, \mathbb{Q}}(x) &= x^2 + x + 1 \\ \mu_{\sqrt{5}, \mathbb{Q}}(x) &= x^2 - 5. \end{aligned}$$

Computing the degree, we observe that $\mu_{\sqrt[3]{11}, \mathbb{Q}(\omega, \sqrt{5})} = \mu_{\sqrt[3]{11}}$, and similarly for $\mu_{\omega, \mathbb{Q}(\sqrt{5})}$ with $\mu_{\omega, \mathbb{Q}}$, so that the degree of $K/\mathbb{Q}$ is 12.

- (b) First, we recall the theorem that a separable polynomial $f(x)$ is irreducible if and only if the action of the Galois group on the roots of $f(x)$ is transitive. Since f is separable (as we are working over $\mathbb{Q}$) and not irreducible, it follows that the Galois group of $f(x)$ is not transitive.

Now, we observe that $f(x)$ has two irreducible factors — namely, $p(x) = x^3 - 11$ (which is irreducible by the Eisenstein criterion) and $q(x) = x^2 + x - 1$, and in particular, the extension $K/\mathbb{Q}$ is equal to the compositum of the splitting fields of $p(x)$ and $q(x)$, which we label as L and M respectively. Since these are finite degree extensions whose intersection is equal to $\mathbb{Q}$, it follows that

$$\text{Gal}(K/\mathbb{Q}) \cong \text{Gal}(L/\mathbb{Q}) \times \text{Gal}(M/\mathbb{Q}).$$

We start by seeing that $L = \mathbb{Q}(\omega, \sqrt[3]{11})$. Since $p(x)$ has degree 3 and not all of its roots are real, it follows that $\text{Gal}(L/\mathbb{Q}) \cong S_3$, while since M has degree 2 over $\mathbb{Q}$, it follows that $\text{Gal}(M/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$, so that $\text{Gal}(K/\mathbb{Q}) \cong S_3 \times \mathbb{Z}/2\mathbb{Z}$.

Problem (Algebra II 2019 Midterm II, Problem 2): Let K/F be a finite Galois extension with $G = \text{Gal}(K/F)$.

- (a) Assume that G is a simple group, with $\alpha \in K \setminus F$ and $\mu_{\alpha, F}(x)$ the minimal polynomial of α over F . Prove that K is a splitting field for $\mu_{\alpha, F}(x)$.
- (b) Let $n = [K : F]$, and fix integers m and ℓ with $m\ell = n$. Find a condition on the subfield lattice of K/F which is equivalent to the following: G can be written as a semidirect product $A \rtimes B$ for some subgroups A and B with $|A| = m$ and $|B| = \ell$.

Solution:

- (a)
- (b) By the definition of the semidirect product, we have that $A \trianglelefteq G$, $G = AB$, and $A \cap B = \{e\}$. In particular, since $|AB| = \frac{|A||B|}{|A \cap B|}$, we have that $G = AB$ is equivalent to $A \cap B = \{e\}$.

We have that $A \trianglelefteq G$ if and only if K^A/F is Galois, that $G = AB$ if and only if $K^{AB} = F$, and that $A \cap B = \{e\}$ if and only if $K^{A \cap B} = K$.

Observe now that $K^{AB} = K^A \cap K^B$. This follows from the fact that $\alpha \in K^{AB} = F$ if and only if $g(\alpha) = \alpha$ for any $g \in AB$, hence for all $g \in A$ and $g \in B$.

Therefore, we claim that $G = A \rtimes B$ with $|A| = m$ and $|B| = \ell$ if and only if K/F contains subfields M_A and M_B such that

- (i) $[M_A : F] = \ell$, $[M_B : F] = m$;

(ii) M_A/F is Galois;

(iii) $M_A \cap M_B = F$.

The forward direction follows from our earlier work by setting $M_A = K^A$ and $M_B = K^B$.